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SUPERSEDING
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MILITARY SPECIFICATION

DISCS, OUTLET VALVE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers natural rubber discs for use as components of protective mask outlet valves.

1.2 Classification. The outlet valve discs shall be of the following types:

Type I – Drawing B5-2-600.

Type II – Drawing B5-2-1006.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-601 – Ruber: Sampling and Testing.

MILITARY

MIL-STD-105 – Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 – Marking for Shipment and Storage.

DRAWINGS

EDGEWOOD ARSENAL

D5-81-3 – Voicemitter - Outlet Valve, C29R1.

B5-2-597 – Valve, Outlet, C15.

B5-2-600 – Disc, Outlet Valve.

B5-2-1006 – Disc, Outlet Valve.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 4240

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification rules should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Rubber compound.* The rubber compound shall be nonblooming and shall contain no less than 92 percent, by volume, of new natural rubber. A slight paraffin bloom shall not be considered a violation of the nonblooming requirement of this specification.

3.1.2 *Antioxidant.* The rubber compound shall contain an antioxidant.

3.1.3 *Copper and manganese inhibitor.* The rubber compound shall contain a copper and a manganese inhibitor.

3.2 Manufacture. The discs shall be manufactured by the conventional process of molding under heat and pressure in accordance with Drawing B5-2-600 for type I or Drawing B5-2-1006 for type II, whichever is applicable.

3.2.1 *Test slabs.* Twenty-five test slabs, 6 by 6 by 0.035 ± 0.012 inches, shall be made from the preproduction sample stock and from the stock used for each regular production lot. The slabs shall have a cure equivalent to that of regular production discs (see 6.3).

3.3 Physical properties.

3.3.1 *Unaged.*

3.3.1.1 *Tensile strength.* The average tensile strength shall be no less than 2600 pounds per square inch.

3.3.1.2 *Ultimate elongation.* The average ultimate elongation shall be no less than 650 percent.

3.3.1.3 *Permanent set.* The average permanent set shall not exceed 15 percent when specimens are stretched 600 percent, held for 10 minutes, released, and rested for 10 minutes.

3.2.2 *Aged (oven process).*

3.3.2.1 *Tensile strength and ultimate elongation.* After aging, the average tensile strength and average ultimate elongation shall increase no more than 20 percent or decrease no more than 15 percent.

3.3.2.2 *Disc characteristics.* After aging, the discs shall exhibit no tackiness or increased hardness. The discs shall not ear when tested as specified in 4.7.4.

3.4 Resistance to airflow. Resistance to airflow through the assembled outlet valve shall not exceed 22 millimeter (mm) of water at an airflow of 85 liters per minute when the outlet valve discs is clean, thoroughly dry, and properly assembled to the outlet valve seat as shown on Edgewood Arsenal Drawing B5-2-597 (C15 Outlet Valve) or D5-81-3 (C29R1 Voicemitter-Outlet Valve), whichever is applicable.

3.5 Leakage. Leakage through the assembled outlet valve shall not exceed 15 ml of air per minute at a pressure droy of 25 mm of water when the outlet valve disc is thoroughly dry and properly assembled to the

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outlet valve seat as shown on Edgewood Arsenal Drawing B5-2-597 (C15 Outlet Valve) or D5-81-3 (C29R1 Voice mitter-Outlet Valve), whichever is applicable.

3.6 **Preproduction sample.** Prior to the start of regular production and following any change in regular production methods or rubber formulation, 100 discs and 25 test slabs (see 3.2.1) shall be produced using the same methods, materials, and equipment as will be used during regular production.

3.7 **Workmanship.** The disc surfaces shall be smooth and free from imperfections such as porous areas, cracks, checks, flash, abrasions, and blisters. The rubber shall be compact and free from holes, bubbles, foreign matter, and tackiness. Ridges of mold parting lines, when closely trimmed, shall not be considered as flash. Rips, tears, or punctures shall be considered as damage.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to specified requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2.3 *Inspection substation.* If the supplier elects to establish substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL-STD-105 are applicable to each substation and to the end item.

4.3 **Preproduction sample inspection.**

4.3.1 *Examination.* Preproduction sample discs shall be inspected for compliance with the drawings and the requirements of this specification.

4.3.2 *Tests.* Twenty-five discs and the test slabs shall be tested in accordance with 4.7 as applicable.

4.3.3 *Acceptance/rejection criteria.* The preproduction sample shall meet the examination and tests specified by 4.3.1 and 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 **Lotting.** A regular production lot shall consist of the outlet valve discs produced by one manufacturer, from one batch of raw material under essentially the same manufacturing conditions.

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4.5 Sampling.

4.5.1 *For examination and nondestructive tests.* Sampling shall be conducted in accordance with MIL-STD-105.

4.5.2 *For destructive tests.*

4.5.2.1 *Physical properties.* Twenty-five test slabs, prepared in accordance with 3.2.2, shall be submitted with each lot for testing.

4.5.2.2 *Disc characteristics.* Twenty-five representative discs shall be taken from each production lot.

4.6 Inspection provisions.

4.6.1 *For examination and nondestructive tests.* Sample discs from regular production lots shall be examined in accordance with the classification of defects and MIL-STD-105.

4.6.2 *For destructive tests.*

4.6.2.1 *Physical properties.* Sample test slabs shall be tested in accordance with 4.7.3. The acceptance number is 0.

4.6.2.2 *Disc characteristics.* Sample discs shall be tested as specified in 4.7.4. The acceptance number is 0.

4.6.3 *Classification of defects.*

4.6.3.1 *Disc, outlet valve, (Dwg B5-2-600 or B5-2-1006).*

Categories	*Defects*	*Acceptance standards*
Critical:	None defined.	
Major:	AQL 1.0 percent defective.	
101	Thickness.	
102	Workmanship (see 3.7).	
#103	Depth (0.182) incorrect.	
Major A:	AQL 0.25 percent defective.	
150	Resistance to airflow.	4.7.5
151	Leakage.	4.7.6
Minor:	AQL 2.5 percent defective.	
201	Marking missing or incorrect.	
202	Stem length (min) incorrect.	

4.7 Tests.

4.7.1 *Test specimens.* Twenty-five specimens conforming to FED-STD-601 shall be cut from the test slabs. The specimens shall not be buffed or stretched before testing. Ten specimens before aging and 10 specimens after aging shall be tested for tensile strength and ultimate elongation. Five specimens shall be tested before aging for permanent set.

4.7.2 *Aging (oven process).* The test specimens selected for oven process aging shall be aged in an air oven at a temperature of 158° ± 2°F. for a continuous period of 168 hours or at a temperature of 212° ± 2°F. for 24 continuous hours, as specified in the applicable sections of FED-STD-601 (see 6.4).

4.7.3 *Tensile strength, ultimate elongation, and permanent set.* Tensile strength,

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ultimate elongation, and permanent set tests of the unaged and aged specimens shall be conducted as specified in the applicable sections of FED-STD-601.

4.7.4 *Disc characteristics.* The sample discs shall be aged as specified in 4.7.2 and examined for tackiness and hardness. Thereafter, the discs shall be tested as follows: Make two gage marks ½ inch apart on the periphery of the disc. Grasp the disc with the fingers at the gage marks, stretch until the gage marks are 1½ inches apart, and permit the disc to return to its original unstressed condition. Examine the disc for tears.

4.7.5 *Resistance to airflow.* The resistance to airflow shall be determined using Edgewood Arsenal Q106 Inlet and Outlet Valve Resistor Indicator.

4.7.6 *Leakage.* Leakage shall be determined using Edgewood Arsenal M4A1 Outlet Valve Leakage Indicator.

5. PREPARATION FOR DELIVERY

(For repair parts packaging see 6.5)

5.1 Packaging, level C. Discs shall be packaged to afford adequate protection from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Discs, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to first domestic destination. Containers shall be in accordance with Uniform Freight Classification rules, or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type I disc is intended for use as a component of the C15 outlet valve. The type II disc is intended for use as a component of the C29R1 outlet valve.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of the spefication.

(b) Type of disc and color.

(c) The valve seats for conducting tests will be furnished by the contracting officer.

6.3 Equivalent cure. Slabs and discs of the same stock and gage are considered to have equivalent cures or the same state of vulcanization when:

(a) The free-sulfur content does not vary more than 10 percent.

(b) The average tensile-stress value at any given elongation (other than ultimate elongation) does not vary more than 2.5 percent.

(c) The curing time can be equated as follows:

$$t_2 = t_1 \times 1.6 \left(\frac{T_1 - T_2}{10} \right)$$

where:

t_1 = minutes of cure at temperature T_1,

t_2 = minutes of cure at temperature T_2,

T_1 = temperature °F. of cure for time t_1, and

T_2 = temperature °F. of cure for time t_2.

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EQUIVALENT CURE TIME
FOR
NATURAL RUBBER

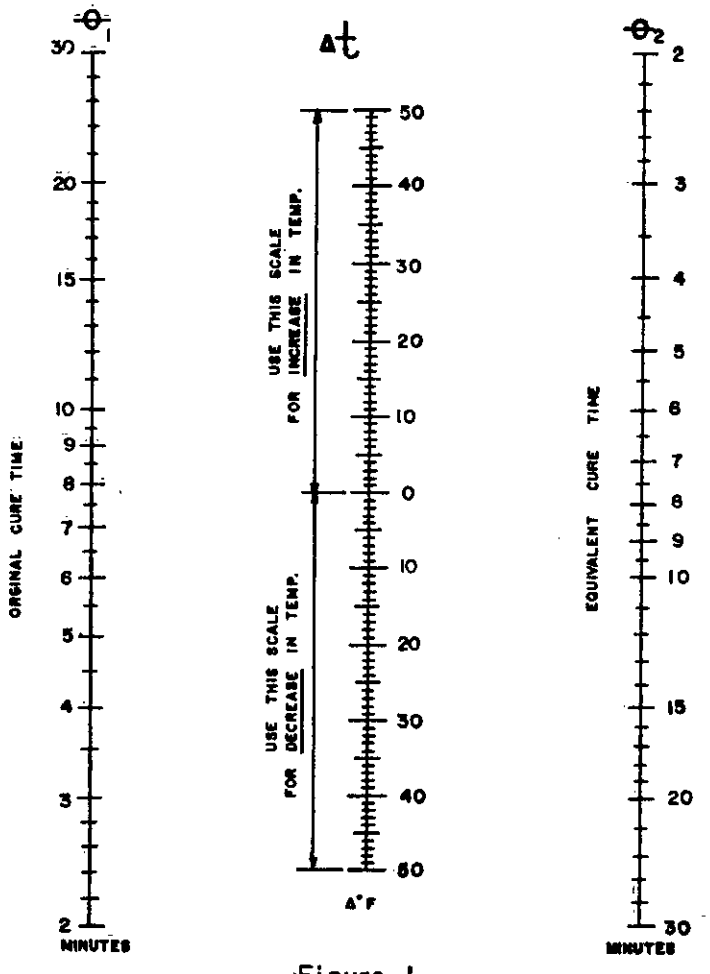


Figure 1.

USING A STRAIGHT EDGE, CONNECT THE ORIGINAL CURE TIME ON θ_1 SCALE WITH THE CHANGE OF CURE TEMPERATURE ON Δt SCALE. EXTEND THE LINE UNTIL IT INTERSECTS θ_2 SCALE AT THE EQUIVALENT CURE TIME.

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For convenience, the nomograph in figure 1 may be used instead of the mathematical computation in determining the equivalent cure.

6.4 **Aging.** Aging at 212° ± 2°F. for a continuous period of 24 hours is slightly more severe on rubber than aging at 158° ± 2°F. for 168 consecutive hours; therefore, specimens aged at the high temperature, that fail to meet requirements may be discarded and replaced by new specimens aged at 158° ± 2°F. for 168 consecutive hours.

6.5 **Repair parts packaging.** When repair parts packaging and packing is required for items authorized for stocking as repair parts, the repair parts packaging data sheets covering the discs should be used. (Copies of repair parts packaging data sheets may be obtained from the Director for Engineering & Industrial Services, ATTN: Chief, Design Branch, U.S. Army Edgewood Arsenal, Edgewood Arsenal, Md. 21010.)

6.6 **International interest.** Certain provisions of this specification are the subject of ABC international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:

Army—MU

Navy—SH

Air Force—84

International interest (see 6.6)

Review activities:

Army—MD

Navy—SH

User activities:

Navy—MC

Air Force—84

Preparing activity:

Army—MU(EA)

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